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⚀⚃ *Roll the dice & look before you leap*: Going beyond the creative limits of next-token prediction

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Abstract

We design a suite of minimal algorithmic tasks that are a loose abstraction of *open-ended* real-world tasks. This allows us to cleanly and controllably quantify the creative limits of the present-day language model. Much like real-world tasks that require a creative, far-sighted leap of thought, our tasks require an implicit, open-ended *stochastic* planning step that either (a) discovers new connections in an abstract knowledge graph (like in wordplay, drawing analogies, or research) or (b) constructs new patterns (like in designing math problems or new proteins). In these tasks, we empirically and conceptually argue how next-token learning is myopic; multi-token approaches, namely teacherless training and diffusion models, comparatively excel in producing diverse and original output. Secondly, to elicit randomness without hurting coherence, we find that injecting noise at the input layer (dubbed *seed-conditioning*) works surprisingly as well as (and in some conditions, better than) temperature sampling from the output layer. Thus, our work offers a principled, minimal test-bed for analyzing open-ended creative skills, and offers new arguments for going beyond next-token learning and temperature sampling. We make part of the code available under `https://github.com/chenwu98/algorithmic-creativity`

1. Introduction

Not all forms of intelligence are solely about being correct or wrong. In *open-ended* tasks, what also matters is finding creative ways to satisfy a request, making surprising and fresh connections never seen before. For instance, consider responding to highly under-specified prompts like "`Generate a challenging high-school word problem involving the Pythagoras Theorem.`" or "`Suggest some candidate therapeutic antibodies targeting the HER2 antigen.`" or "`Provide a vivid analogy to differentiate quantum and classical mechanics.`" Creativity in such tasks requires generating responses that are not just *correct* or *coherent*, but are also *diverse* across responses and are *original* compared to the training data. These currently-sidelined desiderata will rise to prominence as we explore LLMs for open-ended scientific discovery (Gruver et al., 2023; Romera-Paredes et al., 2024; Hayes et al., 2025), for generating novel training data (Yu et al., 2024; Yang et al., 2024c; Wang et al., 2023), and as we scale up test-time compute approaches that benefit from diversity in exploration, such as best-of-N (Cobbe et al., 2021; Chow et al., 2024; Dang et al., 2025) and long chain-of-thought reasoning (OpenAI, 2024; DeepSeek-AI, 2025; Snell et al., 2024; Wu et al., 2024).

Unlike simple open-ended tasks like generating names and basic sentences (Zhang et al., 2024b; Hopkins et al., 2023; Bigelow et al., 2024), many creative tasks (like designing a clever Olympiad problem) are said to involve a random flash of creative insight termed variously as a *leap of thought* (Wang et al., 2024a; Talmor et al., 2020; Zhong et al., 2024), a "eureka" moment (Bubeck et al., 2023), a mental leap (Holyoak & Thagard, 1995; Callaway, 2013; Hofstadter, 1995) or an incubation step (Varshney et al., 2019). The thesis of this paper is that learning to solve such creative leap-of-thought tasks (defined shortly) is misaligned with the current language modeling paradigm (a) in terms of next-token learning, and (b) in how randomness is elicited. We articulate these two concerns by designing a suite of algorithmic tasks inspired by such creative tasks. We then demonstrate how the creativity of language models suffers in these tasks, and how this can be alleviated (to an extent, within our tasks).

Concretely, for the scope of this paper, a creative leap-of-thought task refers to tasks that involve a search-and-plan process; crucially, this process orchestrates multiple *random*

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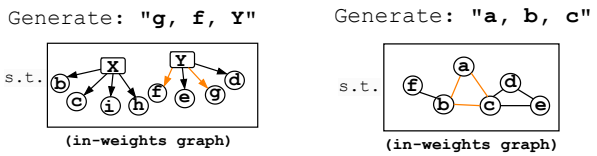
decisions *in advance* before generating the output. Typically, such a leap of thought is *highly implicit* in the text — to infer it, one has to deeply engage with the text and detect higher-order patterns. We can think of tasks like designing satisfying math problems, generating worthwhile research ideas, or drawing surprising analogies as examples of such tasks.

Ideally, one would directly study these real-world tasks to quantify the limits of language models. Indeed, a flurry of recent works report that LLM-generated research ideas tend to be rephrased from existing ideas (Gupta & Pruthi, 2025; Beel et al., 2025) and that LLM outputs tend to be less creative than humans e.g., Chakrabarty et al. (2024); Lu et al. (2024b) (See §J). While assessing real-world tasks is a lofty goal, the evaluations are subjective (Wang et al., 2024b; Runco & Jaeger, 2012), and when the model has been exposed to all of the internet, originality is hard to ascertain. Thus, the conclusions will inevitably invite debate (such as Si et al. (2024) vs. Gupta & Pruthi (2025) or Lu et al. (2024a) vs. Beel et al. (2025)).

In search of more definitive conclusions, we approach from a different angle: we study controllable tasks that are loose abstractions of real-world tasks and yet allow one to objectively quantify originality and diversity. This follows recent works that have studied diversity of models in path-finding (Khona et al., 2024) and challenging context-free grammars (CFGs) (Allen-Zhu & Li, 2023b). We broadly term these as open-ended algorithmic tasks. Our aim then is design minimal instances of these tasks, akin to basic arithmetic for reasoning, so as to tease apart the bare minimum computational skills required for creativity. Then, we can examine how the current modeling paradigm may be limited even with this basic skills, and what alternative paradigms may be desired.

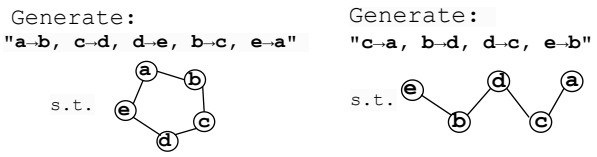
As our first main contribution, we draw inspiration from cognitive science literature (Boden, 2003) (see also Franceschelli & Musolesi (2023)) to design algorithmic tasks isolating two distinct types of creative leaps of thought. The first class of tasks involves *combinational creativity*: drawing novel connections in knowledge, like in research, wordplay or drawing analogies (see Fig 1 for task description). The second class of tasks involves *exploratory creativity*: constructing fresh patterns subject to certain rules, like in designing problems and suspense (see Fig 2). In these tasks, we can precisely evaluate models for the fraction of generations that are *coherent, unique and original* (not present in training set). We term this metric "algorithmic creativity" to denote that it solely evaluates the computational aspects of creativity.

Within this framework, we articulate two creative limits of the current language modeling paradigm. First, we empirically find that next-token learning achieves lower algorithmic creativity compared to some multi-token approach, namely, either teacherless training (Bachmann & Nagarajan, 2024; Monea et al., 2023; Tschannen et al., 2023) or diffusion (Hoogeboom et al., 2021; Austin et al., 2021; Lou et al., 2023) (see Fig 3 and Fig 4). Our argument is that in all our tasks, inferring the latent leap of thought requires observing global higher-order patterns rather than local next-token patterns in the sequence.



(a) Sibling Discovery (b) Triangle Discovery

Figure 1. **Minimal tasks inspired by combinational creativity:** Skills like research, humor and analogies often require identifying novel multi-hop connections from *known* pair-wise relationships in a knowledge graph. For instance, creating the wordplay "What kind of shoes do spies wear? Sneakers." requires searching over a semantic graph, and carefully planning a pair of words (shoes, spies) that lead to a mutual neighbor (sneakers). Inspired by this, we define tasks where a symbolic graph is stored in the model weights; the model is exposed to example node sequences that form a specific multi-hop structure (like a sibling or a triangle) during training. The model must infer this structure from training; during inference, the model must implicitly recall-search-and-plan to generate novel and diverse node sequences obeying the same structure in the in-weights graph. Pictured are two example tasks with a symbolic graph each, and a corresponding example sequence obeying a sibling (g, f, Y) or a triangle structure (a, b, c). More details in §2.3 and Fig 9.



(a) Circle Construction (b) Line Construction

Figure 2. **Minimal tasks inspired by exploratory creativity:** Skills like designing problem sets, novel proteins and plots require devising patterns that can be *resolved* in novel ways through some general rules. Inspired by this, we design a task where during training, we expose the model to "adjacency lists" that implicitly resolve into a specific structure (a circle or a line graph) under some permutation. The model must infer this higher-order structure; during inference, the model must generate adjacency lists resolving to the same structure, but under novel and diverse permutations. Pictured are example sequences and the corresponding implicit structure they would resolve to. See §2.4 and Fig10.

Next, we turn to how we elicit randomness from a Transformer. While the de facto approach is to elicit randomness

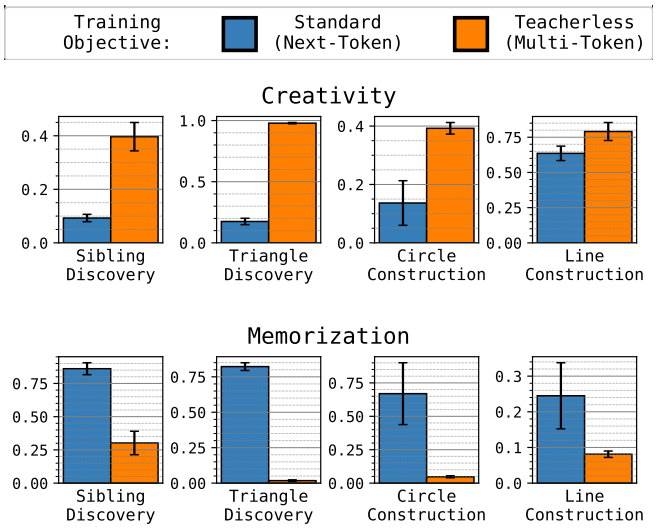


Figure 3. **Multi-token teacherless finetuning improves algorithmic creativity (top; Eq 1) and reduces memorization (bottom; fraction of generations seen during training) on our four open-ended algorithmic tasks for a `Gemma v1 (2B)` model**.

at the output — temperature sampling — we contrast this with injecting randomness into the model. Concretely, we study *seed-conditioning*, where we train and test with a random prefix string per example. Surprisingly, we find that seed-conditioning induces non-trivial algorithmic creativity even with deterministic decoding; it is in fact, comparable to temperature sampling (and in some cases, better). We intuit that maximizing diversity at the output-level results in cognitive overload: it requires simultaneously processing many leaps of thoughts to compute a marginalized token distribution. Input-level noise injection could sidestep this by articulating one leap of thought per seed. We also view this exploration as a controlled study solidifying prior indications that prompt variations induce diversity (Li et al., 2023; Lau et al., 2024; Naik et al., 2024; Li et al., 2022).

Overall, we hope to advance the field in two directions. First, we provide a new angle to advocate for multi-token approaches, orthogonal to the "path-star" example in Bachmann & Nagarajan (2024) (or B&N'24 in short). Whereas, the path-star example portrays a gap in correctness of reasoning, ours shows a gap in diversity of open-ended thinking. We note though that B&N'24 is an impossibility result where next-token learning breaks down spectacularly (unless there is exponential data or compute), while ours is a data-inefficiency result (where next-token learning occurs but is mediocre). Next, the gap we show appears even in 2-token-lookahead tasks as against the many-token-lookahead path-star task. Third, and most conceptually important is the fact that, while the path-star task is amenable to next-token prediction upon reversing the tokens, we identify tasks where no re-ordering is friendly towards next-token prediction — *the optimal thing to do is to globally learn higher-order patterns implicit in the whole future sequence*. This presents a challenge to recent proposals that aim to fix next-token prediction via permutations (Pannatier et al., 2024; Thankaraj et al., 2025) or partial lookaheads (Bavarian et al., 2022; Fried et al., 2022; Kitouni et al., 2024; Nolte et al., 2024).

As a second direction of progress, we hope our work provides a foundation to think about open-ended tasks which are extremely hard to quantify in the wild. This may spur more algorithmic explorations on improving diversity (such as our approach of seed-conditioning) and on curbing verbatim memorization in language models.

Our contributions:

1. We create minimal, controlled and easy-to-quantify open-ended algorithmic tasks. These tasks isolate, and loosely capture two fundamental modes of creativity.
2. We find that multi-token prediction through one of teacherless training or diffusion, results in significantly increased algorithmic creativity in our tasks compared to next-token prediction.
3. Our argument provides new support for multi-token prediction, going beyond B&N'24. We show a gap in creativity in an open-ended task (rather than correctness in a deterministic one), in much simpler 2-token-lookahead tasks, and in tasks where no token permutation is friendly to next-token-learning.
4. We find that input-randomization via seed-conditioning achieves algorithmic creativity comparable with conventional output-randomization via temperature sampling.

2. Open-ended algorithmic tasks & two types of creativity

We are interested in designing simple algorithmic tasks that are loosely inspired by endeavors such as generating scientific ideas, wordplay, narration, or problem-set design, where one needs to generate strings that are both "interesting" and never seen before. In all these tasks, before generating the output, one requires *a (creative) leap of thought*, a process that (a) is implicit i.e., is not spelled out in token space (or is even inherently hard to spell out), (b) involves discrete *random* choices (c) and together, those choices must be *coherent* in that they are carefully planned to satisfy various non-trivial, discrete constraints. These constraints fundamentally define the task and make it interesting e.g., a word problem should be solvable by arithmetic rules, or a pun must deliver a surprising punchline. The goal in such open-ended tasks is not just coherence though, but also diversity and novelty — generations must be as varied as possible and must not be regurgitated training data.